

Python Basics

Variables in python

In [1]:

```
num = 100
str1 = "demo"
print(num)
print(str1)
```

```
100
demo
```

Python multiple assignment

In [2]:

```
x = y = z = 10
print(x)
print(y)
print(z)
```

```
10
10
10
```

Another example of multiple assignment

In [3]:

```
a, b, c = 1, 6, 70
print(a)
print(b)
print(c)
```

```
1
6
70
```

In [4]:

```
X=2
Y=3
X+Y
```

Out[4]:

```
5
```

In [5]:

```
X=2
Y="HELLO"
X+Y
```

TypeError

Traceback (most recent call last)

Input In [5], in <cell line: 3>()

```
1 X=2
2 Y="HELLO"
----> 3 X+Y
```

TypeError: unsupported operand type(s) for +: 'int' and 'str'

Data Types

Python's built-in core data types are in some cases also called object types. There are four built-in data types for numbers:

Number Data Type

- python
- Normal integers: e.g. 4321
- Octal literals (base 8): A number prefixed by a 0 (zero) will be interpreted as an octal number e.g. 0o10
- Hexadecimal literals (base 16): Hexadecimal literals have to be prefixed either by "0x" or "0X". e.g. 0xA0F
- Long integers: these numbers are of unlimited size e.g. 4200000000000000000L
- Floating-point numbers: e.g. 42.11, 3.1415e-10
- Complex numbers: Complex numbers are written as + j x = 3 + 4j

In [6]:

```
a=123
print(a)
b=0o10
print(b)
c=0xA0F
print(c)
d=4200000000000000
print(d)
a1=3+4j
print(a1)
print(a1.real)
print(a1.imag)
```

```
123
8
2575
4200000000000000
(3+4j)
3.0
4.0
```

Python Boolean

These are the decisive data types, they only return categorical value, i.e True or False.

type function

To know type of variable type function is used

In [7]:

```
a="asdsa"  
print(type(a))  
<class 'str'>
```

Implicit and Explicit Data Type Conversion

Implicit conversion or coercion is when data type conversion takes place either during compilation or during run time and is handled directly by Python

In [8]:

```
a_int = 1  
b_float = 1.0  
c_sum = a_int + b_float  
print(c_sum)  
print(type(c_sum))  
2.0  
<class 'float'>
```

Explicit Type Conversion

One can convert from one type to another with the `int()`, `float()`, and `complex()` methods:

`type(var)`

In [9]:

```
x = 1 # int
y = 2.8 # float
z = 1j # complex
a = float(x)
b = int(y)
c = complex(x)
print(a)
print(b)
print(c)

print(type(a))
print(type(b))
print(type(c))
```

```
1.0
2
(1+0j)
<class 'float'>
<class 'int'>
<class 'complex'>
```

In [10]:

```
print("hello world")
```

```
hello world
```

Types of Operator

Python language supports the following types of operators -

- Arithmetic Operator
- Comparison (Relational) Operators
- Assignment Operators
- Logical Operators
- Bitwise Operators
- Membership Operators
- Identity Operators

Arithmetic Operator like +,-,, /,%,*,//

In [11]:

```
a=11
b=2
print(a**b) #exponent
print(a//b) # floor division
```

```
121
5
```

Comparison (Relational) Operators, bitwise operator and assignment operators are same as C programming

Logical Operators are and, or and not

e.g. a and b

Membership Operators

In [12]:

```
l=[1,2,3]
print(1 in l)
print(4 not in l)
```

True

True

Identity Operators

Identity operators compare the memory locations of two objects. Evaluates to true if the variables on either side of the operator point to the same object and false otherwise.

In [13]:

```
x=2
y=2
print(x is y)
z=22
print(x is z)
```

True

False

input function and formatted output

Python input() function syntax input([prompt]) Here the prompt argument is optional. The prompt argument is used to display a message to the user. For example, the prompt is, "Please enter a value." When input() function executes, program flow stops until a user enters some value. Whatever you enter as input, the input() function converts it into a string. If you enter an integer value, still it will convert it into a string.

In [14]:

```
number = input("Enter number ")
name = input("Enter name ")

print("\n")
print("Printing type of a input value")
print("type of number", type(number))
print("type of name", type(name))
```

Enter number 2
Enter name khyati

Printing type of a input value
type of number <class 'str'>
type of name <class 'str'>

In [15]:

```
number = int(input("Enter number "))
name = input("Enter name ")

print("\n")
print("Printing type of a input value")
print("type of number", type(number))
print("type of name", type(name))
print("number={0},name={1}".format(number,name))
print("name={1},numner={0}".format(number,name))
print('number=%4d,name=%s'%(number,name))
```

Enter number 2
Enter name khyati

Printing type of a input value
type of number <class 'int'>
type of name <class 'str'>
number=2,name=khyati
name=khyati,numner=2
number= 2,name=khyati

In [16]:

%history

```
num = 100
str1 = "demo"
print(num)
print(str1)
x = y = z = 10
print(x)
print(y)
print(z)
a, b, c = 1, 6, 70
print(a)
print(b)
print(c)
X=2
Y=3
X+Y
X=2
Y="HELLO"
X+Y
a=123
print(a)
b=0o10
print(b)
c=0xA0F
print(c)
d=4200000000000000
print(d)
a1=3+4j
print(a1)
print(a1.real)
print(a1.imag)
a="asdsa"
print(type(a))
a_int = 1
b_float = 1.0
c_sum = a_int + b_float
print(c_sum)
print(type(c_sum))
x = 1 # int
y = 2.8 # float
z = 1j # complex
a = float(x)
b = int(y)
c = complex(x)
print(a)
print(b)
print(c)

print(type(a))
print(type(b))
print(type(c))
print("hello world")
a=11
b=2
print(a**b) #exponent
print(a//b) # floor division
l=[1,2,3]
print(1 in l)
```

```
print(4 not in 1)
x=2
y=2
print(x is y)
z=22
print(x is z)
number = input("Enter number ")
name = input("Enter name ")

print("\n")
print("Printing type of a input value")
print("type of number", type(number))
print("type of name", type(name))
number = int(input("Enter number "))
name = input("Enter name ")

print("\n")
print("Printing type of a input value")
print("type of number", type(number))
print("type of name", type(name))
print("number={0},name={1}".format(number,name))
print("name={1},numner={0}".format(number,name))
print('number=%4d,name=%s'%(number,name))
%history
```

In [17]:

```
%history 4-6
```

```
X=2
Y=3
X+Y
X=2
Y="HELLO"
X+Y
a=123
print(a)
b=0o10
print(b)
c=0xA0F
print(c)
d=4200000000000000
print(d)
a1=3+4j
print(a1)
print(a1.real)
print(a1.imag)
```

In []: